

Project: GigForce

Contract & Gig Workforce Management Platform

Domain: HR & Workforce

1. Introduction

GigForce is a web-based contract and gig workforce management platform designed for enterprises, staffing agencies, and marketplace operators to manage the full lifecycle of contract workers, freelancers, and project-based talent including sourcing, onboarding, assignment management, timesheets, invoicing, compliance, and workforce analytics. The system enables contractors to manage their profiles, log timesheets, and track payments; hiring managers to raise resource requests and approve timesheets; vendor managers to oversee supplier contracts and worker deployments; and administrators to configure rate cards, compliance rules, and approval workflows.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Contractor / Freelancer:** Manages profile, logs timesheets, tracks assignments, and views payment status
- **Hiring Manager:** Raises resource requests, reviews profiles, and approves timesheets
- **Vendor / Staffing Agency:** Submits worker profiles against requisitions and manages deployments
- **Vendor Manager:** Oversees supplier contracts, performance scorecards, and rate negotiations
- **Finance Executive:** Processes contractor invoices, manages purchase orders, and tracks accruals
- **Workforce Admin:** Configures skill catalogs, rate cards, compliance rules, and approval workflows

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.

- **2.2 Contractor Profile & Skill Management:** Manages contractor profiles, skill tags, certifications, and engagement history.
- **2.3 Resource Requisition & Vendor Sourcing:** Manages resource requests, vendor submissions, shortlisting, and selection workflows.
- **2.4 Assignment & Contract Management:** Manages work assignments, statements of work, extensions, and early terminations.
- **2.5 Timesheet & Leave Management:** Handles timesheet submission, approval workflows, and contractor absence tracking.
- **2.6 Invoicing & Payment Processing:** Manages contractor invoices, purchase orders, accruals, and payment tracking.
- **2.7 Workforce Analytics & Reporting:** Provides dashboards for contractor headcount, spend analysis, and vendor performance.
- **2.8 Notifications & Alerts:** Assignment start/end alerts, timesheet approval reminders, invoice status updates, and compliance expiry warnings.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Contractor, vendor, and enterprise staff registration with role-based access control
- Organisation-scoped session management and full audit logging

Entities

- **User**
 - UserID
 - Name
 - Role (Contractor/HiringManager/Vendor/VendorManager/Finance/Admin)
 - Email

- Phone
- OrgUnitID
- Status (Active/Inactive/Suspended)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - Timestamp

4.2 Contractor Profile & Skill Management Module

Features

- Maintain comprehensive contractor profiles with skills, certifications, and availability
- Track engagement history, performance ratings, and compliance document validity

Entities

- **ContractorProfile**
 - ContractorID
 - UserID
 - DisplayName
 - PrimarySkill
 - SkillTags
 - ExperienceYears
 - PreferredEngagementType (Remote/Onsite/Hybrid)
 - AvailabilityStatus (Available/OnAssignment/OnNotice)
 - Status (Active/Inactive/Blacklisted)
- **ContractorCertification**
 - CertID
 - ContractorID

- CertificationName
- IssuingBody
- IssuedDate
- ExpiryDate
- Status (Valid/Expired/Revoked)
- **EngagementHistory**
 - HistoryID
 - ContractorID
 - ClientName
 - Role
 - StartDate
 - EndDate
 - PerformanceRating
 - FeedbackSummary

4.3 Resource Requisition & Vendor Sourcing Module

Features

- Raise resource requisitions with skill, rate, and duration requirements
- Invite vendors to submit profiles and manage shortlisting and selection workflows

Entities

- **ResourceRequisition**
 - RequisitionID
 - HiringManagerID
 - BusinessUnitID
 - SkillRequired
 - ExperienceLevel (Junior/Mid/Senior)
 - EngagementType (Remote/Onsite/Hybrid)
 - StartDate

- Duration
- MaxRate
- Status (Draft/Open/InProgress/Filled/Cancelled)
- **VendorSubmission**
 - SubmissionID
 - RequisitionID
 - VendorID
 - ContractorID
 - ProposedRate
 - SubmissionDate
 - Status (Submitted/Shortlisted/InterviewScheduled/Selected/Rejected)

4.4 Assignment & Contract Management Module

Features

- Create assignments with statement of work, agreed rates, and engagement terms
- Manage assignment extensions, scope changes, and early termination workflows

Entities

- **Assignment**
 - AssignmentID
 - RequisitionID
 - ContractorID
 - HiringManagerID
 - VendorID
 - StartDate
 - EndDate
 - AgreedRatePerDay
 - EngagementType
 - SOWReference

- Status (Active/Extended/Completed/TerminatedEarly)
- **AssignmentAmendment**
 - AmendmentID
 - AssignmentID
 - AmendmentType (Extension/RateRevision/ScopeChange/EarlyTermination)
 - EffectiveDate
 - NewValue
 - ApprovedByID
 - Status (Pending/Approved/Rejected)

4.5 Timesheet & Leave Management Module

Features

- Allow contractors to log daily or weekly timesheets against active assignments
- Manage multi-level timesheet approval and contractor absence requests

Entities

- **Timesheet**
 - TimesheetID
 - AssignmentID
 - ContractorID
 - WeekStartDate
 - WeekEndDate
 - HoursLogged
 - OvertimeHours
 - SubmittedDate
 - Status (Draft/Submitted/Approved/Rejected/Revised)
- **TimesheetLine**
 - LineID
 - TimesheetID

- Date
- HoursWorked
- ActivityDescription
- Status
- **ContractorAbsence**
 - AbsenceID
 - ContractorID
 - AssignmentID
 - StartDate
 - EndDate
 - Reason
 - ApprovedByID
 - Status (Pending/Approved/Rejected)

4.6 Invoicing & Payment Processing Module

Features

- Generate contractor invoices from approved timesheets against purchase orders
- Track invoice approval, payment processing, and accrual reconciliation

Entities

- **PurchaseOrder**
 - POID
 - AssignmentID
 - VendorID
 - POAmount
 - Currency
 - IssuedDate
 - ExpiryDate
 - Status (Active/Exhausted/Cancelled)

- **ContractorInvoice**
 - InvoiceID
 - POID
 - AssignmentID
 - ContractorID
 - InvoicePeriod
 - HoursBilled
 - InvoiceAmount
 - SubmittedDate
 - Status (Submitted/Approved/Rejected/Paid/Disputed)
- **Payment**
 - PaymentID
 - InvoiceID
 - PaidAmount
 - PaymentDate
 - PaymentMode (BankTransfer/Cheque)
 - Status (Processed/Pending/Failed)

4.7 Workforce Analytics & Reporting Module

Features

- Active contractor headcount, spend by business unit and skill category dashboards
- Vendor performance scorecards, fill rate, time-to-fill, and compliance expiry tracking reports

Entities

- **WorkforceReport**
 - ReportID
 - Scope (BusinessUnit/Vendor/Skill/Period)
 - Metrics (ActiveContractors, TotalSpend, FillRate, AvgTimeToFill, TimesheetApprovalRate, ComplianceExpiryCount)

- GeneratedDate

4.8 Notifications & Alerts Module

Features

- Timesheet submission reminders, approval status updates, and invoice payment notifications for contractors
- Assignment end warnings, certification expiry alerts, and PO exhaustion warnings for managers and vendors

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Assignment/Timesheet/Invoice/Compliance/Vendor)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, OrgUnitID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, Timestamp)
- **ContractorProfile**(ContractorID, UserID, DisplayName, PrimarySkill, SkillTags, ExperienceYears, PreferredEngagementType, AvailabilityStatus, Status)
- **ContractorCertification**(CertID, ContractorID, CertificationName, IssuingBody, IssuedDate, ExpiryDate, Status)

- **EngagementHistory**(HistoryID, ContractorID, ClientName, Role, StartDate, EndDate, PerformanceRating, FeedbackSummary)
- **ResourceRequisition**(RequisitionID, HiringManagerID, BusinessUnitID, SkillRequired, ExperienceLevel, EngagementType, StartDate, Duration, MaxRate, Status)
- **VendorSubmission**(SubmissionID, RequisitionID, VendorID, ContractorID, ProposedRate, SubmissionDate, Status)
- **Assignment**(AssignmentID, RequisitionID, ContractorID, HiringManagerID, VendorID, StartDate, EndDate, AgreedRatePerDay, EngagementType, SOWReference, Status)
- **AssignmentAmendment**(AmendmentID, AssignmentID, AmendmentType, EffectiveDate, NewValue, ApprovedByID, Status)
- **Timesheet**(TimesheetID, AssignmentID, ContractorID, WeekStartDate, WeekEndDate, HoursLogged, OvertimeHours, SubmittedDate, Status)
- **TimesheetLine**(LineID, TimesheetID, Date, HoursWorked, ActivityDescription, Status)
- **ContractorAbsence**(AbsenceID, ContractorID, AssignmentID, StartDate, EndDate, Reason, ApprovedByID, Status)
- **PurchaseOrder**(POID, AssignmentID, VendorID, POAmount, Currency, IssuedDate, ExpiryDate, Status)
- **ContractorInvoice**(InvoiceID, POID, AssignmentID, ContractorID, InvoicePeriod, HoursBilled, InvoiceAmount, SubmittedDate, Status)
- **Payment**(PaymentID, InvoiceID, PaidAmount, PaymentDate, PaymentMode, Status)
- **WorkforceReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Contractor Portal:** Profile manager, active assignments, timesheet entry, invoice tracker, payment history
- **Hiring Manager Console:** Requisition builder, submission review, shortlisting dashboard, timesheet approval queue
- **Vendor Portal:** Submission form, contractor deployment list, invoice submission, PO tracker
- **Vendor Manager Dashboard:** Supplier scorecard, contract registry, rate card management, performance trend charts

- **Finance Dashboard:** PO tracker, invoice approval queue, accrual reconciliation, payment log
- **Admin Console:** Skill catalog, rate card configuration, approval workflow builder, compliance rule setup

8. Non-Functional Requirements

- **Performance:** Support 15,000+ concurrent contractors, vendors, and enterprise users across global deployments
- **Security:** Contractor PII protection, rate and financial data encryption, RBAC, and audit trails for all approval actions
- **Scalability:** Multi-entity and multi-currency support for global enterprises with regional contractor pools
- **Availability:** 99.9% uptime with timesheet submission and invoice approval workflows prioritised for high availability
- **Maintainability:** Configurable rate cards, approval chains, compliance document rules, and engagement templates
- **Observability:** Real-time contractor deployment dashboards, spend analytics, and compliance expiry monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with external payroll systems, ERP purchase order systems, and background verification bureaus
- Invoice payments are recorded as internal status updates; bank transfer API integration is deferred to Phase 2
- All notifications are in-app only; email digest and calendar integrations are out of Phase 1 scope
- Background check results are entered manually; third-party screening API integration is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling